

Finite groups and rings generating varieties of rapid growth

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Abstract

Let A be a finite universal algebra. Then the cardinalities of the n -generated free algebras F_n in the variety (equational class) generated by A satisfy G. Birkhoff's inequality: $|F_n| \leq |A|^{|A|^n}$ for $n = 1, 2, \dots$. It follows that $\limsup_{n \rightarrow \infty} \sqrt[n]{\log |F_n|} \leq |A|$. When A is a finite group or a finite nonassociative algebra, we obtain a criterion for equality in this estimate; equivalently, a criterion for maximal growth of the sequence $\{|F_n|\}_{n=1}^\infty$.

Key words: finite groups, finite rings, varieties of groups and rings.

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1 Introduction

Let A be a finite algebra of a fixed signature. The variety of algebras $\text{var } A$ generated by A consists of all algebras of the same signature satisfying every identity that holds in A . A basic theorem of G. Birkhoff [3, 5] says that $\text{var } A$ is the smallest class of algebras containing A and closed under taking subalgebras, homomorphic images, and Cartesian products.

The cardinality of every n -generated algebra in the variety $\text{var } A$ is bounded by the cardinality of the n -generated free algebra F_n in $\text{var } A$. The algebra F_n isomorphically embeds in a direct power of the algebra A , and its cardinality is bounded from above by the double-exponential function $|A|^{|A|^n}$, where $|A|$ is the cardinality of the algebra A (see [3] or subsection 0.3 in [5]). It follows that $\limsup_{n \rightarrow \infty} \sqrt[n]{\log |F_n|} \leq |A|$. We denote the left-hand side of this inequality by $\text{pot } A$ and call it the *potential* of the finite algebra A .

For example, if G is a nontrivial finite group, then $\text{pot } G = 1$ if G is nilpotent and $\text{pot } G \geq 2$ otherwise ([7], 24.52). The same alternative holds for finite-dimensional (non-associative) algebras over finite fields [1].

As early as 1937, it was noticed by B.H. Neumann [6] that Birkhoff's estimate can be improved in many cases. Our goal is to find necessary and sufficient conditions on A under which the potential $\text{pot } A$ attains its maximal possible value, namely conditions equivalent to the equality $\text{pot } A = |A|$, where A is either a finite group or a finite, not necessarily associative, ring.

Recall that a nontrivial group (ring) A is called *monolithic* if the intersection of all nontrivial normal subgroups (2-sided ideals) of A is nontrivial. This intersection is the *monolith* of the monolithic group (respectively, ring) A .

Theorem 1. *Let G be a finite group. The equality $\text{pot } G = |G|$ holds if and only if*
() G is a monolithic group with a non-abelian monolith.*

Moreover, under condition (), there exist positive constants c_1 and c_2 such that $c_1|G|^n < \log |F_n| < c_2|G|^n$ for the n -generated free groups F_n in the variety $\text{var } G$.*

Many nonsimple groups satisfy condition (*); for example, the symmetric groups S_n for $n \geq 5$ and any wreath product of a non-cyclic finite simple group and a nontrivial finite group.

A (not necessarily associative) non-zero ring A is a *prime ring* if the product of any two nonzero (2-sided) ideals of A is nonzero. The following is an analogue of Theorem 1.

Theorem 2. *Let A be a (non-associative) finite ring. The equality $\text{pot } A = |A|$ holds if and only if A is a prime ring.*

Moreover, if A is a prime ring, then there exist positive constants c_1 and c_2 such that $c_1|A|^n < \log |F_n| < c_2|A|^n$ for the n -generated free rings F_n in the variety $\text{var } A$.

Classical results in ring theory imply that every finite prime associative ring is simple ([9], Section 98). By contrast, the following construction gives a nonsimple prime algebra over a field with basis (e, f) and multiplication rules $e^2 = f$, $ef = f$, $fe = 0$, and $f^2 = f$. It has only one one-dimensional ideal $\langle f \rangle$.

2 Sufficiency in Theorem 1

Lemma 1. *Let G be a finite monolithic group with a non-abelian monolith. Let H be a direct product $H_1 \times \dots \times H_k$ with projections $\pi_i : H \rightarrow H_i$. Let S be a subdirect product, i.e. a subgroup of H such that $\pi_i(S) = H_i$ for all i , with an epimorphism $\phi : S \rightarrow G$. Then there exist an index i and an epimorphism $\psi : H_i \rightarrow G$ such that $\phi = \psi\pi_i$.*

Proof. By induction on k ; this is trivial when $k \leq 1$; assume $k \geq 2$. The subgroups $\phi(S \cap H_i)$ are normal and pairwise centralize each other, so at most one of them is nontrivial since the monolith of $G = \phi(S)$ is non-abelian. After reindexing, we may assume $\phi(S \cap H_k)$ is trivial. Let π be the projection $S \rightarrow H_1 \times \dots \times H_{k-1}$ and $S' = \pi(S)$. Then $\phi\pi^{-1}$ is a well-defined epimorphism $S' \rightarrow G$ since $\ker \pi = S \cap H_k$ and $\phi(S \cap H_k) = \{1\}$. By induction, there exists $i < k$ and an epimorphism $\psi : H_i \rightarrow G$ such that $\phi\pi^{-1} = \psi\pi'_i$, where π'_i is the projection of $H_1 \times \dots \times H_{k-1}$ onto H_i . Hence $\phi = \psi\pi'_i\pi = \psi\pi_i$, as required. $\square$

Lemma 2. *Let G be a finite monolithic group with non-abelian monolith M . Let G^I be a finite direct power of G , with direct factors G_i isomorphic to G for $i \in I$. For a fixed $n \geq 1$, let $x_1, \dots, x_n \in G^I$ and $x_{1i}, \dots, x_{ni}$ be their projections in G_i . Assume that the elements $x_{1i}, \dots, x_{ni}$ generate G_i for $i \in I$ and suppose the n -tuples $(x_{1i}, \dots, x_{ni})$, $i \in I$, lie in pairwise distinct $(\text{Aut } G)$ -orbits on G^n . Then the subgroup B of G^I generated by $x_1, \dots, x_n$ contains M^I .*

Proof. Arguing by contradiction, assume that B does not contain the copy M_i of the monolith M for some $i \in I$. Since M_i is the monolith of G_i , it follows that $B \cap M_i = \{1\}$, because the subgroup $B \cap M_i$ of the subdirect product B must be normal in the factor G_i . Therefore $B \cap G_i = \{1\}$ since nontrivial normal subgroups of G_i contain M_i . Hence B can be subdirectly embedded in the product $\prod_{j \neq i} G_j$.

By Lemma 1, the projection $\pi_i : B \rightarrow G_i$ factors as $\psi\pi_j$, where $j \neq i$ and ψ is an isomorphism $G_j \rightarrow G_i$. Thus, the n -tuple $(x_1, \dots, x_n)$ is mapped to $(x_{1i}, \dots, x_{ni})$ under π_i and it is mapped to $(x_{1j}, \dots, x_{nj})$ under ψ . It follows that the isomorphism ψ maps $(x_{1j}, \dots, x_{nj})$ to $(x_{1i}, \dots, x_{ni})$, contrary to the assumption that these n -tuples belong to different $\text{Aut } G$ -orbits. The lemma is proved by contradiction. $\square$

Remark 1. For finite simple, non-abelian groups G , the statement of Lemma 2 has been known for a long time, see [4].

For each proper subgroup $H \leq G$, we have $|H| \leq |G|/2$ and the number of n -tuples of elements from H is equal to $|H|^n$. If s is the number of proper subgroups of G , then the number of n -tuples generating G is at least $|G|^n - s(|G|/2)^n = |G|^n(1 - O(2^{-n}))$. Thus, for some constant $c > 0$ and all sufficiently large n , there are at least $\frac{|G|^n}{|\text{Aut } G|} (1 - O(2^{-n})) > c|G|^n$ n -tuples lying in pairwise distinct $(\text{Aut } G)$ -orbits. By Lemma 2, there is an n -generated subgroup $B = B(n)$ of order at least $|M|^{c|G|^n}$ in some G^I . Since $B(n) \in \text{var } G$, we obtain $\log |F_n| \geq \log |B(n)| > c_1|G|^n$ for some $c_1 > 0$ and every $n \geq 1$, as required. Birkhoff's inequality implies $\log |F_n| < c_2|G|^n$ for some $c_2 > 0$. This proves the sufficiency part of Theorem 1.

3 Necessity in Theorem 1

Lemma 3. Let G_1 and G_2 be finite groups. Assume that a finite group G generates the variety $\mathfrak{V} = \text{var } (G_1, G_2)$. Then $\text{pot } G = \max(\text{pot } G_1, \text{pot } G_2)$. In particular, if G is not monolithic, then $\text{pot } G = \text{pot } (G/N) \leq |G/N|$ for a non-trivial normal subgroup N of G .

Proof. Clearly, $\text{pot } G \geq \max(\text{pot } G_1, \text{pot } G_2)$. On the other hand, the n -generated $\mathfrak{V}$ -free group F_n is a subdirect product of the free groups F'_n and F''_n in the varieties $\text{var } G_1$ and $\text{var } G_2$ ([7], 15.82), and hence $|F_n| \leq |F'_n||F''_n|$. To estimate $\text{pot } G$ from above, it remains only to note that, for any sequences of positive integers $\{a'_n\}_{n=1}^\infty$ and $\{a''_n\}_{n=1}^\infty$, we have

$$\begin{aligned} \limsup_{n \rightarrow \infty} \sqrt[n]{\log(a'_n a''_n)} &= \limsup_{n \rightarrow \infty} \sqrt[n]{\log a'_n + \log a''_n} \\ &= \max(\limsup_{n \rightarrow \infty} \sqrt[n]{\log a'_n}, \limsup_{n \rightarrow \infty} \sqrt[n]{\log a''_n}). \end{aligned}$$

If the group G is not monolithic, there are two nontrivial normal subgroups N_1 and N_2 in G such that $N_1 \cap N_2 = \{1\}$. Then G is a subdirect product of the factor groups $G_1 = G/N_1$ and $G_2 = G/N_2$, and $\text{var } G = \text{var } \{G_1, G_2\}$. Hence $\text{pot } G = \max(\text{pot } G_1, \text{pot } G_2)$, which completes the proof. $\square$

Lemma 4. Let Q be a group and N be a (right) Q -module. Assume that $N = \oplus_i M_i$ is a direct sum of isomorphic finite Q -submodules M_i of cardinality m . Then any n -generated submodule L of N has cardinality at most m^{nm} .

Proof. There are module isomorphisms $\alpha_{ij} : M_i \rightarrow M_j$ such that $\alpha_{jk}\alpha_{ij} = \alpha_{ik}$ for all i, j, k . If $a \in M_i$ and the indices $i, i_1, \dots, i_k$ are pairwise different, then the 'diagonal-type' sum $b = a + \alpha_{ii_1}(a) + \dots + \alpha_{ii_k}(a)$ generates a Q -submodule isomorphic to the submodule generated by a , and so b generates a submodule of order at most m . Since every element a' of M_j equals $\alpha_{ij}(a)$ for some $a \in M_i$, any element of N can be decomposed as a sum of at most m 'diagonal' elements. Hence every one-generated submodule of N has order at most m^m , and so $|L| \leq (m^m)^n$, as required. $\square$

Lemma 5. Let G be a finite group and N an abelian normal subgroup of a group $H \in \text{var } G$, on which the finite group $R = H/N$ acts by conjugation, and $N = \oplus_i M_i$ be a direct sum of isomorphic

finite R -submodules M_i of cardinality m . Then for every $n \geq 1$, any n -generated subgroup S of H has order at most $|G|^{|G|^{\log_2 |R|}} m^{nm}$.

Proof. One can choose at most $\log_2 |R|$ generators, say $x_1, \dots, x_k$, in S whose natural images generate the subgroup $Q = SN/N \leq R$. By multiplying the other $n - k$ generators by some products of $x_1^{\pm 1}, \dots, x_k^{\pm 1}$, one obtains a set of generators $\{x_1, \dots, x_k, y_{k+1}, \dots, y_n\}$ in S , where $y_i \in N$ for $i > k$.

Since the subgroup K generated by $x_1, \dots, x_k$ belongs to $\text{var } G$, the order of K does not exceed $C = |G|^{|G|^{\log_2 |R|}}$. The normal closure L of the elements $y_{k+1}, \dots, y_n$ in S is a Q -submodule and so its cardinality is at most $m^{(n-k)m}$ by Lemma 4. Thus, $|S| = |KL| \leq |K||L| \leq Cm^{nm}$. $\square$

Proposition 1. *If M is an abelian normal subgroup of a finite group G , then $\text{pot } G \leq |G/M|$.*

Proof. Recall that the n -generated $(\text{var } G)$ -free group F_n is embedded in a direct power G^I for some I . Denote by H the product $F_n M^I$, so H/M^I is a subgroup of T^I , where $T = G/M$. For $i \in I$, let $y_{1i}, \dots, y_{ni}$ be the projections of the free generators $x_1, \dots, x_n$ of F_n in the factor G_i , and $z_{1i}, \dots, z_{ni}$ be their natural images in the quotients G_i/M_i . We call two indices i and j equivalent if the n -tuples $(z_{1i}, \dots, z_{ni})$ and $(z_{1j}, \dots, z_{nj})$ are equal (if we identify both T_i and T_j with T). So the set I is a disjoint union of at most $|T|^n$ equivalence classes.

For an equivalence class J , we denote by H_J the projection of H to the direct power G^J . It follows from the definition of the equivalence that the projections of the n -tuple of the free generators of F_n to G_j are equal modulo M^J for all $j \in J$, whence the image of F_n in G^J/M^J is contained in the diagonal of $(G/M)^J$ and the quotient $R = H_J/M^J$ has order at most $|T|^n$. Moreover, the abelian normal subgroups M_j of H_J are isomorphic R -modules for all $j \in J$.

By Lemma 4, the order of the projection S_J of F_n to H_J does not exceed $|G|^{|G|^{\log_2 |T|}} |M|^{n|M|}$. Since the number of equivalence classes J is at most $|T|^n$, and the group F_n is contained in the direct product of subgroups H_J over all J , we have

$$|F_n| \leq (|G|^{|G|^{\log_2 |T|}} |M|^{n|M|})^{|T|^n}.$$

The logarithm of the right-hand side is bounded from above by $cn|T|^n$, where the constant c does not depend on n . It follows that $\text{pot } G \leq |T|$, as required. $\square$

The statement of Theorem 1 follows from the following

Proposition 2. *(Gap for the values of $\text{pot } G$). Let G be a non-trivial finite group. Then $\text{pot } G = |G|$ if G is a monolithic group whose monolith is non-abelian. Otherwise $\text{pot } G \leq d < |G|$ for some divisor d of $|G|$.*

Proof. The first statement of the proposition is proved in Section 2, the second one follows from Lemma 3 and Proposition 1. $\square$

4 Proof of Theorem 2

Note that a finite prime ring A cannot contain two different minimal ideals, and multiplication is nonzero on the unique minimal ideal. If, for brevity, rings with zero multiplication are called abelian (as in Lie theory), A is a monolithic ring with a non-abelian monolith M .

Such a reformulation makes Theorem 2 very similar to Theorem 1, and the proofs are similar as well. Of course, the variety $\text{var } A$ contains rings, not groups; one works with subrings instead of subgroups and with ideals instead of normal subgroups. If the expression 'centralize each other' is replaced by 'annihilate each other on both sides' in the proof of Lemma 1, we obtain the formulation and proof of an analogous Lemma 1' for rings.

The obvious changes in vocabulary are also sufficient for the formulation and proof of a Lemma 2' analogous to Lemma 2. (Of course, one refers to Lemma 1' instead of Lemma 1 in the proof.) The last paragraph of Section 2 can be trivially reformulated as well. The same changes lead to the formulation and proof of Lemma 3.

Let R be a ring, and N be an abelian ideal of R . Then for $a \in N$ and $r \in R$, the products ra and ar depend only on the coset $r + N$, and so the left and right multiplications of a by the elements of R/N are well defined. Given two abelian ideals M and M' in R , we say that these ideals are *similar in R* if there is an isomorphism of their additive groups $\alpha : M \rightarrow M'$ such that $\alpha(rm) = r\alpha(m)$ and $\alpha(mr) = \alpha(m)r$ for any $m \in M$ and $r \in R$. Clearly, the similarity of ideals is an equivalence relation.

Lemma 4'. *Let R be a ring, and let N be an ideal of R that is a direct sum $\oplus_i M_i$ of finite abelian ideals M_i of cardinality m that are similar in R . Then any n elements of N are contained in an ideal L of R of cardinality at most m^{nm} .*

The proof follows that of Lemma 4, where α_{ij} is an R -similarity of the ideals M_i and M_j , and the minimal ideals containing $a \in M_i$ and the 'diagonal' element b (as there) are similar in R .

Lemma 5'. *Let A be a finite ring and N an abelian ideal of a ring $H \in \text{var } A$, $N = \oplus_i M_i$ be a direct sum of similar in H ideals M_i of cardinality m , and $R = H/N$. Then for every $n \geq 1$, any n -generated subring S of H has cardinality at most $|A|^{|A|^{\log_2 |R|}} m^{nm}$.*

For the proof of Lemma 5', one changes the vocabulary (in particular, the words 'isomorphic modules' are replaced by 'ideals similar in G '). In the proof, the element y_i ($i > k$) is obtained as a sum of x_i and an element from the subring K generated by $x_1, \dots, x_k$. Now L is the minimal ideal of S containing $y_{k+1}, \dots, y_n$, and $S = K + L$.

Only obvious changes are needed in the formulations and proofs of Propositions 1' and 2'. This completes the proof of Theorem 2. The author believes that similar modifications would lead to the proof of an analogue of Theorem 2 for Ω -rings.

Remark 2. *Let A be a finite-dimensional (non-associative) algebra over a finite field $\mathbf{k}$, and let F_n denote the n -generated free algebra in the variety $\text{var } A$ of algebras over $\mathbf{k}$ generated by A . Then one can formulate an analogue of Theorem 2 and prove it without any new assumptions. In this setting, however, it is preferable to consider the dimension function $d(n) = \dim_{\mathbf{k}} F_n$, which is the logarithm of the cardinality function $|F_n|$. Thus the potential $\text{pot } A$ of A is $\limsup_{n \rightarrow \infty} \sqrt[n]{d(n)}$, and Theorem 2 is modified as follows:*

Let A be a finite-dimensional (nonassociative) algebra over a finite field $\mathbf{k}$. The equality $\text{pot } A = |A|$ holds if and only if A is a prime algebra.

Moreover, if A is a prime algebra, then there exist positive constants c_1 and c_2 such that $c_1|A|^n < \dim_{\mathbf{k}} F_n < c_2|A|^n$ for the n -generated free algebras F_n in the variety $\text{var } A$.

5 Examples and questions

It is worth finding potentials for various classes of finite groups and rings. By Theorem 1 and Lemma 3, it suffices to examine monolithic groups and rings having an abelian monolith.

Example 1. Let G be a finite dihedral group. If it is a 2-group, then $\text{pot } G = 1$, because G is nilpotent. Otherwise $\text{pot } G \geq 2$ as mentioned in the Introduction. On the other hand, G has an abelian subgroup of index 2, and therefore $\text{pot } G = 2$ in this case by Proposition 1.

Example 2. Let G be a Miller–Moreno group, that is a non-abelian finite group, all of whose proper subgroups are abelian. We assume that G is not nilpotent, since otherwise $\text{pot } G = 1$. Then, for two distinct prime numbers p and q , $G = PQ$, where P is an elementary abelian p -group, Q is a cyclic q -group, and Q^q lies in the center of G (see [8], 9.1.9). So any non-abelian monolithic quotient $\bar{G}$ of G belongs to the variety $\mathfrak{V}$ consisting of groups having a normal elementary abelian p -subgroup with an elementary abelian q -group as a quotient. Then $\bar{G}$ generates the variety $\mathfrak{V}$ (see [7], 54.41). The order of the free group F_n in $\mathfrak{V}$ is equal to $q^n p^{(n-1)q^n+1}$ ([7], 21.13), whence $\text{pot } G = \text{pot } \bar{G} = q$. For example, $\text{pot } A_4 = 3$ for the alternating group A_4 of degree 4.

Example 3. The following product of varieties of associative algebras (no 1 in the signature) over a field was introduced in the paper of G. Bergman and J. Lewin [2]. The product $\mathfrak{U}\mathfrak{V}$ of two varieties is defined by the set of all identities $u(x_1, \dots, x_k)v(x_{k+1}, \dots, x_{k+\ell}) = 0$, where $u(x_1, \dots, x_k) = 0$ is an identity of the variety $\mathfrak{U}$ and $v(x_1, \dots, x_\ell) = 0$ is an identity of $\mathfrak{V}$. Assume that $\mathfrak{U} = \text{var } A$ and $\mathfrak{V} = \text{var } B$, where A and B are non-zero finite associative algebras over a finite field $\mathbf{k}$. Then, by Proposition 18 and Theorem 20 of [1], the variety $\mathfrak{U}\mathfrak{V}$ is generated by a finite algebra C over $\mathbf{k}$ such that $\text{pot } C = (\text{pot } A)(\text{pot } B)$.

Example 4. Given a finite field $\mathbf{k}$, there is only one non-abelian 2-dimensional Lie algebra L over $\mathbf{k}$ up to isomorphism. It has basis (e, f) such that $ef = -fe = f$ and $e^2 = f^2 = 0$. There is a 1-dimensional monolith $\langle f \rangle$, and $\text{pot } L \leq |L/\langle f \rangle| = |\mathbf{k}|$ by Proposition 1'. To obtain the opposite inequality, we construct an n -generated algebra A in $\text{var } L$ as follows.

Consider the set V^* of $|\mathbf{k}|^{n-1}$ linear functions $\alpha : V \rightarrow \mathbf{k}$, where V is an $n - 1$ -dimensional abelian Lie algebra over $\mathbf{k}$. For each $\alpha \in V^*$, let $\langle f_\alpha \rangle$ be a 1-dimensional Lie algebra and define the Lie module $M_\alpha = \langle f_\alpha \rangle$ over V by the rule $vf_\alpha = \alpha(v)f_\alpha$ for $v \in V$. These rules define the semidirect sum A of V and the ideal $I = \bigoplus_{\alpha \in V^*} M_\alpha$. The ideal I is generated by one element $\sum_{\alpha} f_\alpha$ since the weights α are pairwise distinct. So A is an n -generated Lie algebra of dimension $n + |\mathbf{k}|^{n-1}$. It is embedded in a direct power of L whence $A \in \text{var } L$. (For each α , consider a homomorphism that sends v to $\alpha(v)e$, sends f_α to f , and kills the other weight spaces.) Since $\lim_{n \rightarrow \infty} \sqrt[n]{n + |\mathbf{k}|^{n-1}} = |\mathbf{k}|$, we have $\text{pot } L \geq |\mathbf{k}|$.

Question 1. Is the potential $\text{pot } G$ an integer for every finite group G ?

Question 2. Does $\lim_{n \rightarrow \infty} \sqrt[n]{\log |F_n|}$ exist for any finite group G , where F_n are the n -generated free groups in the variety $\text{var } G$?

Question 3. In the notation of Question 2, does $\lim_{n \rightarrow \infty} \frac{\log |F_{n+1}|}{\log |F_n|}$ exist? (Then it is equal to the limit from Question 2 and to $\text{pot } G$.)

Similar questions for finite algebras have been formulated in [1].

If a finite group G has a central subgroup Z and the quotient G/Z is simple (in particular, if G is quasisimple), then $\text{pot } G = \text{pot } (G/Z)$ by Theorem 1 and Proposition 1. The equality $\text{pot } G = \text{pot } (G/Z) = 1$ also holds if G/Z is a non-trivial nilpotent group. So I venture to ask

Question 4. Let Z be a proper subgroup (subring) of a finite group (ring) G that centralizes (resp., annihilates) G . Does the equality $\text{pot } G = \text{pot } (G/Z)$ hold?

If a group G has an abelian normal subgroup N and $Q = G/N$, there is a semidirect product $\bar{G} = QN$, $Q \cap N = \{1\}$, such that the action by conjugation of Q on N coincides in G and in $\bar{G}$. The following construction (depending on N) shows that $\bar{G} = \bar{G}(N)$ belongs to $\text{var } G$.

Let H be the direct product $G_1 \times G_2$ of two copies of G and D be the diagonal of G . By definition, $K = DN_1 = DN_2$, where N_1 and N_2 are copies of N . Note that $D \cap N_1 = D \cap N_2 = \{1\}$ and $L = D \cap (N_1 \times N_2)$ is the diagonal of $N_1 \times N_2$. It is a normal subgroup of K since N is abelian. Denote the quotient K/L by $\bar{G}$. Then the image Q of D in $K/L = \bar{G}$ has trivial intersection with the image of N_1 ($=$ image of N_2) in $\bar{G}$, and $\bar{G}$ is the required semidirect product.

The same construction defines the semidirect sum $\bar{G} \in \text{var } G$ if G is a ring and N is an abelian ideal of G . This leads to the following generalization of Question 4.

Question 5. Let N be an abelian normal subgroup (ideal) in a finite group (resp., ring) G . In the above notation, does the equality $\text{pot } G = \text{pot } (\bar{G})$ hold?

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